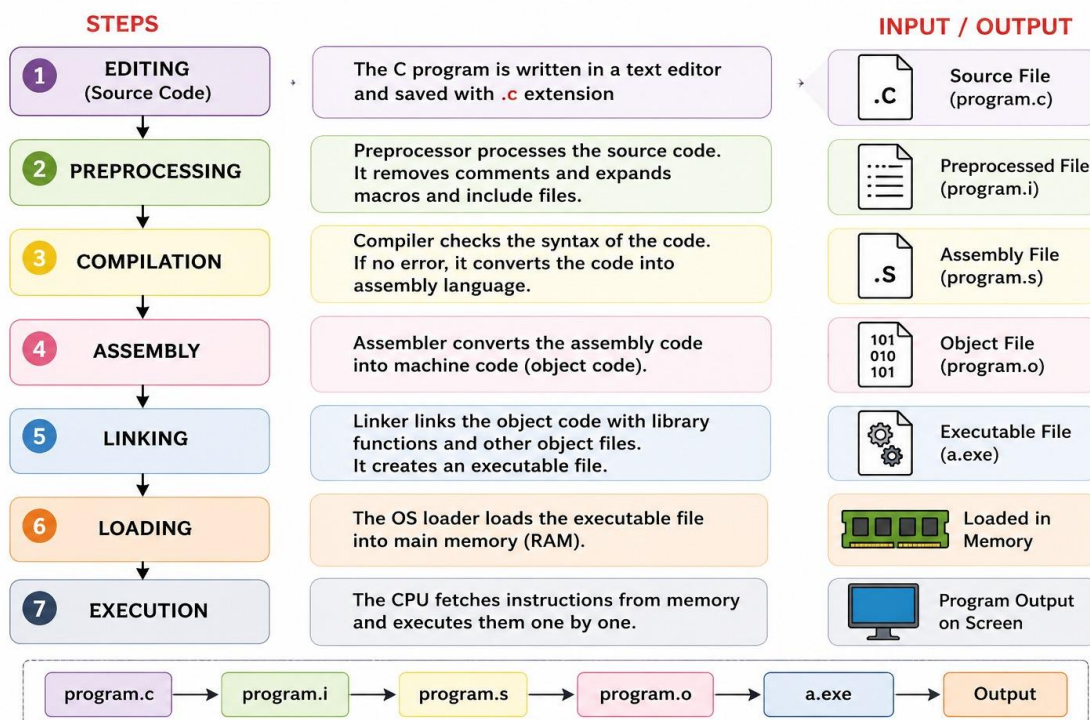


Programming Languages

Course Content

Unit	Language	Topics	Weightage
1.	C	Programming Basics (C): - Structure of a C program; compilation and execution - Data types, variables, constants, operators (arithmetic, relational, logical) - Input/output — scanf/printf in C - Type conversion; simple expressions <i>Real connect: C powers embedded systems, microcontrollers, and OS kernels.</i>	20%
2.	C	Decision Making and Control Flow (C): - Control Statements - Loops — for, while, do-while; break and continue - Functions — definition, call, return; call by value	20%

EXECUTION STEPS OF A C PROGRAM



Data Types, Variables, Constants and Operators in C

1. Data Types

A **data type** specifies the type of data that a variable can store.

Basic Data Types in C

Data Type	Meaning	Example
int	Integer numbers	10, -5, 100
float	Decimal numbers	10.5, 3.14
double	Large/precise decimal numbers	12.345678
char	Single character	'A', 'b', '5'

Examples

```
int age = 18;
```

```
float marks = 85.5;
```

```
double price = 1250.5678;
```

```
char grade = 'A';
```

2. Variables

A **variable** is a named memory location used to store a value that can be changed during program execution.

Syntax

```
data_type variable_name;
```

Example

```
int age;
```

```
float marks;
```

```
char grade;
```

Variable Initialization

Assigning a value to a variable at the time of declaration.

```
int age = 18;
```

```
float marks = 85.5;
```

```
char grade = 'A';
```

Changing the Value

```
int age = 18;
```

```
age = 20;
```

The value of age changes from 18 to 20.

Rules for Naming Variables

1. Variable name can contain **letters, digits and underscore** (_).
2. It cannot start with a digit.
3. Spaces are not allowed.
4. C keywords cannot be used as variable names.
5. C is **case-sensitive**.

Valid

```
age
```

```
student_age
```

```
marks1
```

```
total_marks
```

Invalid

```
1age
```

```
student age
```

```
float
```

3. Constants

A **constant** is a value that does not change during program execution.

Types of Constants

Integer Constant

10

25

-50

Floating-Point Constant

3.14

10.5

-2.5

Character Constant

'A'

'b'

'5'

String Constant

"Hello"

"Computer Science"

Using const

The const keyword is used to create a variable whose value cannot be changed.

```
const int MAX = 100;
```

```
const float PI = 3.14;
```

Example:

```
const int x = 10;
```

```
x = 20; // Error
```

4. Operators

An **operator** is a symbol used to perform an operation on one or more values.

Example:

$a + b$

Here, $+$ is an operator.

5. Arithmetic Operators

Arithmetic operators are used to perform mathematical calculations.

Operator	Operation	Example
$+$	Addition	$a + b$
$-$	Subtraction	$a - b$
$*$	Multiplication	$a * b$
$/$	Division	a / b
$\%$	Remainder (Modulus)	$a \% b$

Example

```
int a = 10, b = 3;
```

```
printf("%d", a + b); // 13
```

```
printf("%d", a - b); // 7
```

```
printf("%d", a * b); // 30
```

```
printf("%d", a / b); // 3
```

```
printf("%d", a \% b); // 1
```

Important Point

For integer division:

$10 / 3 = 3$

The decimal part is discarded.

6. Relational Operators

Relational operators are used to **compare two values**.

The result is either **true (1)** or **false (0)**.

Operator	Meaning	Example
==	Equal to	a == b
!=	Not equal to	a != b
>	Greater than	a > b
<	Less than	a < b
>=	Greater than or equal to	a >= b
<=	Less than or equal to	a <= b

Example

```
int a = 10, b = 20;
```

```
printf("%d", a < b);
```

Output: 1

Because $10 < 20$ is true.

Important Difference

```
a = 10;    // Assignment
```

```
a == 10;   // Comparison
```

= is an **assignment operator**, while == is a **relational operator**.

7. Logical Operators

Logical operators are used to combine or reverse conditions.

Operator	Name	Meaning
&&	Logical AND	Both conditions must be true

Operator	Name	Meaning
!	Logical NOT	Reverses the result

Logical AND &&

(a > 5 && b > 5)

True only when **both conditions are true**.

Logical OR ||

(a > 5 || b > 5)

True when **at least one condition is true**.

Logical NOT !

!(a > 5)

Reverses the result.

Truth Table

A	B	A && B	A B
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

NOT Truth Table

A	!A
0	1

A	!A
1	0

Input/Output in C — scanf() and printf()

1. scanf() Function

scanf() is a **formatted input function** used to read data from the **standard input device (keyboard)**.

It is defined in the header file:

```
#include <stdio.h>
```

Syntax

```
scanf("format_string", argument_list);
```

Example

```
int a;
```

```
scanf("%d", &a);
```

Here, %d is the format specifier for an integer and &a represents the address of variable a.

Reading Multiple Variables

```
int a, b, c;
```

```
scanf("%d %d %d", &a, &b, &c);
```

2. printf() Function

printf() is a **formatted output function** used to display data on the **standard output device (console)**.

It is defined in:

```
#include <stdio.h>
```

Syntax

```
printf("format_string", argument_list);
```

Example

```
int a = 10;
```

```
printf("%d", a);
```

3. Format Specifiers

Format specifiers specify the **type and format of data** to be read or displayed.

Format Specifier	Data Type / Use
%d	Integer
%i	Integer
%u	Unsigned integer
%f	Floating-point number
%lf	Double (scanf)
%c	Single character
%s	String
%o	Octal integer
%x	Hexadecimal integer
%X	Hexadecimal integer (uppercase)
%e	Scientific notation
%g	Fixed or scientific notation

4. scanf() with Different Data Types

Integer

```
int a;
```

```
scanf("%d", &a);
```

Float

```
float a;
```

```
scanf("%f", &a);
```

Double

```
double a;
```

```
scanf("%lf", &a);
```

Character

```
char ch;
```

```
scanf("%c", &ch);
```

String

```
char name[20];
```

```
scanf("%s", name);
```

5. printf() with Different Data Types

```
int a = 10;
```

```
float b = 12.5;
```

```
double c = 25.6789;
```

```
char ch = 'A';
```

```
char name[] = "Ankit";
```

```
printf("%d", a);
```

```
printf("%f", b);
```

```
printf("%lf", c);
```

```
printf("%c", ch);
```

```
printf("%s", name);
```

6. Format String

A **format string** contains ordinary characters and format specifiers that determine how input or output is handled.

Example:

```
printf("Age = %d", age);
```

Here:

- "Age = %d" → Format string
- %d → Format specifier
- age → Argument

7. Escape Sequences with printf()

Escape sequences are used to represent special characters.

Escape Sequence	Meaning
\n	New line
\t	Horizontal tab
\b	Backspace
\r	Carriage return
\\	Backslash
\'	Single quotation mark
\"	Double quotation mark

Example

```
printf("Hello\nWorld");
```

Output:

Hello

World

8. Width and Precision in printf()

Field Width

```
printf("%5d", 25);
```

The output occupies a field of width 5.

Precision

```
printf("%.2f", 12.3456);
```

Output:

12.35

Combined

```
printf("%8.2f", 12.3456);
```

Here:

- 8 → Field width
- .2 → Precision
- f → Floating-point value

9. Difference Between scanf() and printf()

scanf()	printf()
Formatted input function	Formatted output function
Takes input from keyboard	Displays output on console
Stores input in variables	Displays values of variables
Generally uses & for variables	Does not use & for normal variables
Example: scanf("%d", &a);	Example: printf("%d", a);

Type Conversion in C

Type conversion is the process of converting a value from **one data type to another data type**.

For example, converting an int value into a float value.

Type Hierarchy in C

For basic arithmetic type conversion, you can remember the following order from **lower** → **higher**:

Level	Data Type	Typical Size
1	char	1 byte
2	short int	2 bytes
3	int	4 bytes
4	long int	4/8 bytes
5	long long int	8 bytes
6	float	4 bytes
7	double	8 bytes
8	long double	8/12/16 bytes

Order to Remember

Char => short => int => long => long long => float => double => long double

Example

```
int a = 10;
```

```
float b = 2.5;
```

```
float c = a + b;
```

Here, int is converted to float before the addition.

Important: This is a simplified teaching hierarchy for common C expressions. “**Higher**” does **not simply mean “larger size”**; C's actual conversion rules depend on the types involved.

There are **two types of type conversion in C**:

1. Implicit Type Conversion

Implicit conversion is performed **automatically by the compiler**.

Example

```
int a = 10;
```

```
float b;
```

```
b = a;
```

Here, the int value 10 is automatically converted into float.

```
a = 10
```

```
b = 10.0
```

Example

```
int a = 10;
```

```
float b = 2.5;
```

```
float result = a + b;
```

Here, a is automatically converted from int to float.

General Rule

Smaller/Lower type → Larger/Higher type

Example:

int → float → double

2. Explicit Type Conversion (Type Casting)

Explicit conversion is performed **manually by the programmer** using a **cast operator**.

Syntax

(data_type) expression

Example

```
int a = 10;
```

```
int b = 3;
```

```
float result;
```

```
result = (float)a / b;
```

```
printf("%f", result);
```

Output:

3.333333

Here:

(float)a convert's a from int to float.

Important Example

Without Type Casting

```
int a = 5, b = 2;
```

```
float result = a / b;
```

```
printf("%f", result);
```

Output:

2.000000

Because both a and b are integers, integer division is performed.

With Type Casting

```
int a = 5, b = 2;
```

```
float result = (float)a / b;
```

```
printf("%f", result);
```

Output:

2.500000

Implicit vs Explicit Conversion

Implicit Conversion	Explicit Conversion
Done automatically	Done by programmer

Implicit Conversion	Explicit Conversion
No cast operator required	Cast operator is required
Performed by compiler	Specified by programmer
Example: float b = a;	Example: float b = (float)a;

Implicit: Compiler converts the type automatically.

Explicit: Programmer tells the compiler which type to convert using:

(data_type)

Simple Expressions in C

An expression in C is a combination of constants, variables and operators that produces a value.

Example

a + b

x * 5 + 10

(a + b) / 2

1. Operator Precedence

Operator Precedence determines which operator is performed first when an expression contains multiple operators.

Example

```
int result = 10 + 5 * 2;
```

Here, * has higher precedence than +.

So:

10 + 5 * 2

= 10 + 10

= 20

Common Precedence Order

Priority	Operators	Description
1	()	Parentheses
2	* / %	Multiplication, Division, Modulus
3	+ -	Addition, Subtraction
4	< <= > >=	Relational
5	== !=	Equality
6	&&	Logical AND
7	?:	Conditional
8	=	Assignment

Higher priority number in the table means lower precedence.

Example

```
int x = 10 + 20 * 3;
```

First:

$$20 * 3 = 60$$

Then:

$$10 + 60 = 70$$

Therefore:

$$x = 70$$

2. Associativity

Associativity determines the direction in which operators of the same precedence are evaluated.

There are two directions:

- Left to Right
- Right to Left

Example: Left-to-Right

`int x = 20 / 5 * 2;`

/ and * have the same precedence.

Their associativity is left to right.

So:

$20 / 5 * 2$

$= 4 * 2$

$= 8$

Not:

$20 / 10 = 2$

Example: Right-to-Left

Assignment operators have right-to-left associativity.

`a = b = c = 10;`

Evaluation:

`c = 10`

`b = c`

`a = b`

Therefore:

`a = 10`

`b = 10`

`c = 10`

Precedence vs Associativity

Precedence	Associativity
Determines which operator is performed first	Determines direction of evaluation
Used when operators have different precedence	Important when operators have the same precedence
Example: * before +	Example: * and / are left-to-right

Precedence → Which operator first?

Associativity → Which direction?

Example

`int x = 10 + 20 * 3 - 5;`

Step 1: * first

$10 + 60 - 5$

Step 2: + and - have same precedence and are left-to-right

$70 - 5$

$= 65$

Answer: 65

When multiplication (*) and division (/) have the same precedence, C follows left-to-right associativity.

So, whichever appears first from the left is executed first.

Example 1

$20 / 5 * 2$

Both / and * have the same precedence.

From left to right:

$$20 / 5 * 2$$

$$= 4 * 2$$

$$= 8$$

Answer = 8

Example 2

$$20 * 5 / 2$$

From left to right:

$$20 * 5 / 2$$

$$= 100 / 2$$

$$= 50$$

Answer = 50

Operator Precedence and Associativity

Operator Precedence

Operator precedence determines **which operator is evaluated first** when an expression contains different operators.

For example:

$$5 + 2 * 3$$

***** has higher precedence than **+**, so:

$$5 + (2 * 3)$$

$$= 5 + 6$$

$$= 11$$

Operator Associativity

Associativity determines the **order of evaluation** when operators have the same precedence.

For example:

$$20 / 5 * 2$$

/ and * have the same precedence and **left-to-right associativity**:

$$(20 / 5) * 2$$

$$= 4 * 2$$

$$= 8$$

Evaluate the Expression (Imp)

Given:

$$x = 5 * 2 - 8 / 4 + 3 > 6 \&\& 1$$

We apply the operators according to their precedence.

Step 1: Multiplication and Division

* and / have higher precedence than + and -.

$$5 * 2 = 10$$

$$8 / 4 = 2$$

Therefore:

$$x = 10 - 2 + 3 > 6 \&\& 1$$

Step 2: Addition and Subtraction

+ and - have the same precedence and are evaluated **left to right**.

$$10 - 2 + 3$$

$$= 8 + 3$$

$$= 11$$

Therefore:

$$x = 11 > 6 \&\& 1$$

Step 3: Relational Operator >

$$11 > 6$$

This is **true**, so in C:

$$11 > 6 = 1$$

Therefore:

$$x = 1 \ \&\& \ 1$$

Step 4: Logical AND &&

$$1 \ \&\& \ 1 = 1$$

Therefore:

Final Answer

$$x = 1$$

Complete Order of Evaluation

$$5 * 2 - 8 / 4 + 3 > 6 \ \&\& \ 1$$

↓

$$10 - 2 + 3 > 6 \ \&\& \ 1$$

↓

$$8 + 3 > 6 \ \&\& \ 1$$

↓

$$11 > 6 \ \&\& \ 1$$

↓

$$1 \ \&\& \ 1$$

↓

$$1$$

Final value: x = 1

TYPE

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int num1 = 2, num2 = 4, num3 = 7;
```

```
    float sum, avg;
```

```
    float num4= 7.5;
```

```
    sum = num1 + num2 + num3 + num4;
```

```
    avg = (float)sum / 3;
```

```
    printf("\nThe Sum of Three Numbers    = %f", sum);
```

```
    printf("\nThe Average of Three Numbers = %.2f\n", avg);
```

```
}
```

```
#include <stdio.h>
```

```
int main() {
```

```
    // Variable declarations
```

```
    float sub1 = 70, sub2= 80, sub3= 50, sub4= 78, sub5= 94;
```

```
    float total, percentage;
```

```
    // Calculate total and percentage
```

```
    total = sub1 + sub2 + sub3 + sub4 + sub5;
```

```
    percentage = (total / 500.0) * 100.0;
```

```
    // Display the results
```

```
printf("\nTotal Marks Obtained: %.2f / 500.00\n", total);

printf("Percentage: %.2f%%\n", percentage);

return 0;

}
```

IF ELSE

```
#include <stdio.h>

int main() {

    int number = 70;

    if (number >= 50) {

        printf("number is an greater than 50.");

    }

    else {

        printf("number is an less than 50.");

    }

    return 0;

}
```

```
#include <stdio.h>

int main() {

    int number = 70;


    if (number == 50) {

        printf("number is an equal to 50.");

    }

    else if (number >= 50)
```

```

    {

        printf("number is an greater than 50.");

    }

    else {

        printf("number is an less than 50.");

    }

    return 0;

}

#include <stdio.h>

int main() {

    int number = 70;

    if (number >= 50) {

        printf("%d is an greater than 50.",number);

    }

    else {

        printf("%d is an less than 50.",number);

    }

    return 0;

}

// Check whether an integer is odd or even

#include <stdio.h>

int main() {

    int number;

```

```
if (number%2 == 0) {  
    printf("%d is an even integer.",number);  
}  
else {  
    printf("%d is an odd integer.",number);  
}  
return 0;  
}  
printf("Enter an integer: ");  
scanf("%d", &number);
```